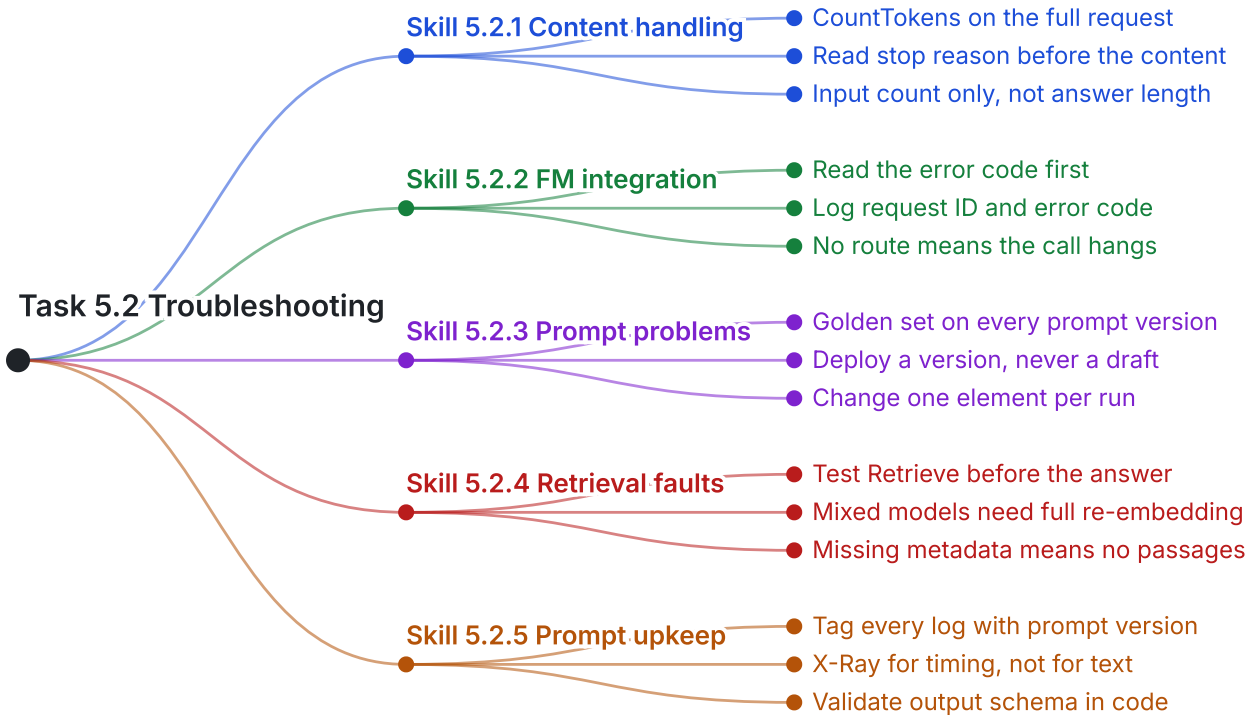
Memory sheet: Task 5.2 Troubleshoot GenAI applications

Domain 5, weight 11 percent. Read this sheet, then answer the flash questions aloud.



Deciding rules

When the stem says	Think of	Not when
Stop reason is max tokens	Output limit, so tighten the format or raise the limit	Stop reason shows a normal end of generation
Answer ends normally but ignores late context	Input side, so count tokens and prune context	The output limit was reached
A call hangs inside a VPC function	Missing NAT route or bedrock-runtime endpoint	AccessDenied comes back quickly
AccessDenied through an interface endpoint	Endpoint policy actions, plus the identity policy	Only the role permissions were widened
The right passage is missing from results	Fix chunking or search before the generator	The passages are right and the answer is wrong
Only some documents were re-embedded	Full corpus re-embedding with one model	Only a prompt wording changed
Throttled at a low request rate with large prompts	Token budget per request and bounded queues	Request count alone explains the throttle
429 ThrottlingException on Bedrock calls	Account quota, so budget tokens and back off	A 503 ServiceUnavailable, which is a shared capacity limit

Etch lines

1. Name the failing layer before you name any fix.
2. The stop reason separates an output limit from a normal end.
3. A longer timeout adds no route, so the call fails the same way.
4. Adding role permissions does not fix an endpoint policy denial.
5. Re-embed the full corpus when the model or preprocessing changes.
6. Underline the symptom words before you read the options.

Flash questions

- Q: The stop reason reads max tokens. Is the fix on the input side or the output side? A: The output side, so tighten the format or raise the configured limit.
- Q: Why does a call with no route wait rather than fail fast? A: The packets have nowhere to go, so the call runs until a timeout ends it.
- Q: Which fields must each prompt log line carry so that failures can be grouped by version? A: The prompt version, with the template ID and validation status.
- Q: An AccessDenied comes back through the endpoint. Which two policies must be checked? A: The endpoint policy and the caller identity policy, since both must allow the call.
- Q: Only the new documents were re-embedded, and answers are outdated. What is the fix? A: Re-embed every document with one model and one preprocessing pipeline, then retest retrieval.
- Q: Which Bedrock error code signals that the account exceeded its quota? A: ThrottlingException, which returns 429.
- Q: Why can flat invocation counts hide a problem during throttling? A: Invocations counts accepted requests only, so check InvocationThrottles beside it.
- Q: A workflow retries a 400 validation error several times. What should happen instead? A: Route it to a catch state at once, since the same input fails on every attempt.

Source: Task 5.2 skill headings 5.2.1 to 5.2.5, Services and techniques in scope, and X versus Y pairs. AWS docs and the study guide sections above, no practice-test text.